GENERATIVE DIRECTED ACYCLIC GRAPH (DAG) OF BAYESIAN MODEL

Hierarchical Formulation for Brain Morphometry under Clinical Slice Degradation & Gadolinium Confounding

HYPERPRIOR

Disease Scale Hyperprior

Controls magnitude of diagnostic phenotype offsets

$\tau_\beta \sim \text{Half-Normal}(0,\ 0.05)$

HYPERPRIOR

Scanner Site Scale Hyperprior

Controls inter-scanner calibration variance across 6 hospitals

$\tau_\gamma \sim \text{Half-Normal}(0,\ 0.05)$

GROUP EFFECT (NON-CENTERED)

Disease Diagnostic Offsets (5 Classes)

Baseline: $\beta_{\text{CONTROL}} \equiv 0$ | Partial pooling for small groups

$z_\beta \sim \mathcal{N}(0,1) \quad \longrightarrow \quad \beta_d = \tau_\beta \cdot z_\beta$

$\beta_d \in \{\text{CONTROL, DEMENTIA, EPILEPSY, HYDROCEPHALUS, PARKINSON}\}$

RANDOM INTERCEPT (NON-CENTERED)

Scanner Site Intercepts (6 Centers)

Absorbs regional B_0 field and RF coil variations

$z_\gamma \sim \mathcal{N}(0,1) \quad \longrightarrow \quad \gamma_s = \tau_\gamma \cdot z_\gamma$

$\gamma_s \in \{\text{RSUTH, UPTH, IDC, AKTH, BMH, LifeBridge}\}$

FIXED PRIOR

Global Baseline

Expected 1mm Control

$\alpha \sim \mathcal{N}(0,1)$

DETERMINISTIC LINK

Linear Predictor (Expected Metric Value per Subject)

Integrates baseline geometry + disease deviation + scanner shift + contrast enhancement

$$\mu_i = \alpha + \beta_{\text{diagnosis}[i]} + \gamma_{\text{site}[i]} + \delta \cdot \text{Contrast}_i$$

FIXED PRIOR

Gadolinium Shift

$\text{Contrast}_i \in \{0, 1\}$

$\delta \cdot \text{Contrast}_i$

$\delta \sim \mathcal{N}(0,\ 0.1)$

Subject Plate: $i \in \{1, \ldots, N\} \quad (N = 218)$

OBSERVED DATA / LIKELIHOOD

Gaussian Likelihood

Evans' Index (EI), Parenchymal Envelope (PEF), or VBR

$$y_i \sim \mathcal{N}(\mu_i,\ \sigma_i^2)$$

Heteroskedastic variance σ_i^2 penalizes low-confidence thick slices

SCALE MODEL (MEASUREMENT NOISE)

Heteroskedastic Dispersion

Exponential slice thickness blur elasticity

$\sigma_i = \sigma_0 \cdot \exp\left(\lambda \cdot [\text{SliceThickness}_i - 1.0]\right)$

$\sigma_0 \sim \text{Half-Normal}(0,\ 0.05) \quad | \quad \lambda \sim \mathcal{N}(0,\ 0.15)$